Yifan Hao

Ph.D. Student in Computer Science | LLM Post-Training & RL & Statistic Machine Learning
yifanh12@illinois.edu | Website | Google Scholar

RESEARCH INTERESTS

My research interests lie at the **intersection of machine learning and statistics**. I am particularly interested in **reinforcement learning for LLM post-training**, **developing efficient algorithms** for modern machine learning systems, and exploring the **theoretical foundations** that explain their effectiveness.

EDUCATION

University of Illinois Urbana-Champaign 2024.9 – present
Ph.D. Student, Computer Science
Advisor: Prof. Tong Zhang

The Hong Kong University of Science and Technology 2022.9 – 2024.8
MPhil, Mathematics
Advisor: Prof. Tong Zhang

University of Science and Technology of China 2018.9 – 2022.6
Bachelor of Science, School of the Gifted Young
Ranking: 5/91 in Statistics

RESEARCH EXPERIENCE

Applied Scientist Intern, Amazon Rufus 2026.5 – 2026.8
Hosts: Zoey Li, Zhenyu Shi

- Developed a **Muon recipe for RLVR** with no momentum. Achieved **29.17%** macro-averaged Pass@1 on Qwen3-4B-Base versus **24.47%** for AdamW across three math benchmarks.
- Used **spectral diagnostics and checkpoint interventions** to analyze optimization stability and identify how Muon organizes learned behavior across singular directions and attention projections.
- Investigated **RLVR trajectory predictability** with random-walk controls, showing that low-rank structure and strong in-sample fits alone do not establish out-of-sample predictability.

Ph.D. Student, University of Illinois Urbana-Champaign 2024.9 – present
Advisor: Prof. Tong Zhang

- Developed **Adaptive Layerwise Perturbation (ALP)**, using learnable perturbations to layer inputs to stabilize off-policy LLM RL under policy staleness and training–inference mismatch, improving mathematical and multi-turn tool-integrated reasoning.
- Proposed the **Distributional Input Projection Network (DIPNeT)**, which demonstrates strong performance across diverse scenarios, including reasoning tasks and adversarial robustness.
- Developed **GVM-RAFT**, a prompt-specific dynamic sample allocation strategy designed to minimize stochastic gradient variance under computational budget constraints, achieving a 2–$4\times$ speedup and significant accuracy improvements over vanilla RAFT.
- Identified and analyzed the **feature decoupling phenomenon** in Transformers, and established theoretical results on their expressiveness for modeling hidden Markov processes.
- Provided a theoretical explanation for the benefits of **ensemble methods** in supervised fine-tuning, attributing their effectiveness to reduced overadaptation.

Master, The Hong Kong University of Science and Technology 2022.9 – 2024.8
Advisor: Prof. Tong Zhang

- Demonstrated the **harmful effects of benign overfitting** on adversarial robustness in overparameterized models, and characterized the trade-off between standard and adversarial risks.
- In the out-of-distribution (OOD) regime, provided theoretical insights into how **spurious feature diversification and overparameterization** improve OOD generalization.

SELECTED PUBLICATIONS AND PREPRINTS

Muon for RLVR: What Works, Why, and What It Learns. [Preprint]
Yifan Hao, Zoey (Sha) Li, Changlong Yu, Qin Lu, Shuowei Jin, Xin Liu, Zhenyu Shi, Tong Zhang.

Adaptive Layerwise Perturbation: Unifying Off-Policy Corrections for LLM RL. [arXiv]
Chenlu Ye*, Xuanchang Zhang*, Yifan Hao*, Zhou Yu, Ziji Zhang, Abhinav Gullapalli, Hao Chen, Jing Huang, Tong Zhang

Towards Better Generalization via Distributional Input Projection Network. [arXiv]
Yifan Hao*, Yanxin Lu*, Hanning Zhang, Xinwei Shen, Tong Zhang

Transformers as Multi-task Learners: Decoupling Features in Hidden Markov Models. [arXiv]
Yifan Hao*, Chenlu Ye*, Chi Han, Tong Zhang

On the Benefits of Over-parameterization for Out-of-Distribution Generalization. [arXiv]
Yifan Hao*, Yong Lin*, Difan Zou and Tong Zhang

The Surprising Harmfulness of Benign Overfitting for Adversarial Robustness. [IEEE Transactions on Information Theory]
Yifan Hao and Tong Zhang

Optimizing Chain-of-Thought Reasoners via Gradient Variance Minimization in Rejection Sampling and RL. [NeurIPS 2025]
Jiarui Yao*, Yifan Hao*, Hanning Zhang, Hanze Dong, Wei Xiong, Nan Jiang, Tong Zhang

Understanding Overadaptation in Supervised Fine-Tuning: The Role of Ensemble Methods. [ICML 2025]
Yifan Hao*, Xingyuan Pan*, Hanning Zhang*, Chenlu Ye, Rui Pan, Tong Zhang

Spurious feature diversification improves out-of-distribution generalization. [ICLR 2024]
Yong Lin*, Lu Tan*, Yifan Hao*, Honam Wong, Hanze Dong, Weizhong Zhang, Yujiu Yang and Tong Zhang

Reverse Diffusion Monte Carlo. [ICLR 2024]
Xunpeng Huang*, Hanze Dong*, Yifan Hao, Yi-An Ma and Tong Zhang.

SFQRA: Scaled Factor-augmented Quantile Regression with Aggregation in Conditional Mean Forecasting. [Journal of Multivariate Analysis]
Lei Shu, Yifan Hao, Yu Chen and Qing Yang

* indicates alphabetical author order or equal contribution.

SELECTED HONORS AND AWARDS

Hong Kong PhD Fellowship (approx. $90,000$ USD over two years)	2022
Outstanding Graduation of Anhui Province	2022
Outstanding Graduation of University of Science and Technology of China	2022
China National Scholarship (top 1.8% by China's Ministry of Education)	2021
Gold Prize for Outstanding Student Scholarship	2020, 2018

TECHNICAL SKILLS

Programming: Python, PyTorch, MATLAB, R, C/C++

TEACHING ASSISTANT EXPERIENCE

CS 441 - Applied Machine Learning	Spring 2026
MATH 2011 - Introduction to Multivariable Calculus	Fall 2023
MATH 2421 - Probability	Spring 2023, Spring 2024
Mathematical Analysis B3	Fall 2021
Complex Analysis	Spring 2021

PROFESSIONAL ACTIVITIES

Conference Reviewer: NeurIPS, ICLR, ICML, AISTATS, AAAI
Journal Reviewer: SIAM Journal on Mathematics of Data Science, JMLR, Transactions on Pattern Analysis and Machine Intelligence